★ GATE/CS 2024

ARRAYS

Date: 12/10/23 | Subject: DSA (Data Structures) ✓Revision 01

① INTRODUCTION & DEFINITION

An array is a collection of homogeneous elements (same data type) stored in contiguous memory locations. Elements can be accessed directly using an index (0 to n-1).

Example: int arr[5] = {10, 20, 30, 40, 50};

Size 'n', Indices '0' to 'n-1', Memory block.

Index :	0	1	2	3	4
	10	20	30	40	50
	1000	1004	1008	1012	1016

Memory block.

② ADDRESS CALCULATION (1D Array)

To find memory address of arr[i] :

$$\boxed{Addr(arr[i]) = BaseAddr + (i * size\_of\_element)}$$

Imp formula for GATE!

Example: Find addr of arr[3] if Base=1000, size=4B.

$1000 + (3 * 4) = 1012$

③ ADVANTAGES (Pros)

1. Direct/Constant Time Access (O(1)) via index. Imp! Recall!
2. Memory efficiency (contiguous storage).
3. Cache friendliness (due to spatial locality).

④ DISADVANTAGES (Cons)

1. Static size (declared at compile time, hard to resize).
2. Inserting/Deleting an element is expensive (O(n)) due to shifting elements.
3. Memory must be contiguous (can cause fragmentation).

⑤ IMPORTANT GATE POINTS ★

Recall!

- Complexity: Access O(1), Insertion/Deletion O(n), Search O(n) or O(log n) (Sorted)
- Multidimensional Arrays: 'row-major' and 'col-major' order (Formulas needed!)
- 'arr[i] is same as *(arr + i)'
- Sizeof(arr) gives total size, Sizeof(arr[0]) gives single element size.

⑥ COMMON QUESTIONS ?

Gate 2019 Q

Q1: Diff between Static vs Dynamic Arrays? (Dynamic: resizing std::vector, ArrayList)

Q2: Matrix multiplication complexity? $O(N^3)$

Q3: How to check if Array is sorted? (Check arr[i] <= arr[i+1] for all i)